ECE 329 Exam 1

Jinhao Zhang

TOTAL POINTS

93.5 / 100

QUESTION 1

35 pts

1.1 a (i) **4 / 4**

✓ **- 0 pts** *Correct*

1.2 a (ii) **4 / 4**

✓ **- 0 pts** *Correct*

1.3 b **5 / 5**

✓ **- 0 pts** *Correct*

1.4 c **4 / 4**

✓ **- 0 pts** *Correct*

1.5 d **5 / 5**

✓ **- 0 pts** *Answer b*

1.6 e **4 / 4**

✓ **- 0 pts** *Answer b*

1.7 f **4 / 4**

✓ **- 0 pts** *Answer c*

1.8 g **5 / 5**

✓ **- 0 pts** *Answer d*

QUESTION 2

10 pts

2.1 a **5 / 5**

✓ **- 0 pts** *Correct*

2.2 b **0.5 / 5**

✓ **- 4.5 pts** *incorrect answer, correct units only*

QUESTION 3

15 pts

3.1 a **5 / 5**

✓ **- 0 pts** *Correct: -40C*

3.2 b **5 / 5**

✓ **- 0 pts** *Correct: 0C*

3.3 c **5 / 5**

✓ **- 0 pts** *Correct: $$\frac{5\sqrt 2}{2} C$$ or $$\frac{5}{\sqrt 2} C$$ or $$\frac{10}{2\sqrt 2} C$$*

QUESTION 4

20 pts

4.1 a **10 / 10**

✓ **- 0 pts** *Correct*

4.2 b **10 / 10**

✓ **- 0 pts** *Correct*

QUESTION 5

20 pts

5.1 a **5 / 5**

✓ **- 0 pts** *Correct*

5.2 b **5 / 5**

✓ **- 0 pts** *Correct*

5.3 c **3 / 5**

✓ **- 2 pts** *Click here to replace this description.*

5.4 d **5 / 5**

✓ **- 0 pts** *Correct (f or bdf)*

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University of Illinois at Urbana-Champaign
ECE 329 Fields and Waves I

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Instructors: Goddard, Ilie, Zhao

September 20, 2023

Name: Jinhao Zhang
Section: X

This exam contains 10 pages (including this cover page) and 5 questions. Total of points is 100. This is a closed book exam and calculators/electronic devices are not allowed. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate. The exam is double sided. You may use the back of the exam as scratch paper. Good luck!

Distribution of Points

Question	Points	Score
1	35	
2	10	
3	15	
4	20	
5	20	
Total:	100	

PART I: MULTIPLE CHOICE QUESTIONS

Circle all that apply! Note that some of these questions require some calculations to get the right answers, but you do not need to show your work for credit. However, partial credit can be given if the work is provided and partially correct.

1. (a) (8 points) A point charge +q located at the origin (0,0,0) of a Cartesian coordinate system is replaced by a spherical conducting shell of radius R carrying the amount of charge +Q and centered on (0,0,0).

   (i) Assuming the zero potential to be at infinity, then the conditions inside the spherical conducting shell can be described as:

   (a) V=0, **E**=0

   (b) V is non-zero and constant, **E** = 0

   (c) V = 0, **E** is non-zero and constant

   (d) V and **E** are both non-zero and constant

   (ii) Now assume that the spherical conducting shell is completely filled with conducting material, the charge on it remaining unchanged, then at points within the sphere

   (a) **E** will be unchanged, but V changed

   (b) **E** will be changed, but V will be unchanged

   (c) **E** and V will both be changed

   (d) **E** and V will remain unchanged

   (b) (5 points) The electric potential at a point (x, y, z) in a certain region of space is given by $V(x, y, z) = \alpha x^2 y$, where α is an arbitrary constant. What is the direction of the electric field at the point (1,0,2)?

   (a) Parallel to Ox.

   (b) Parallel to Oy.

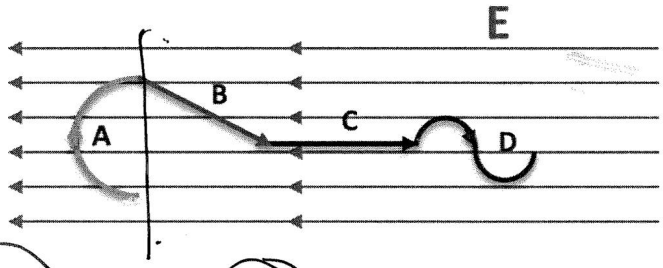
   (c) Parallel to Oz.

   (d) None of the above.

$2\alpha xy \quad \alpha x^2 \quad 0$

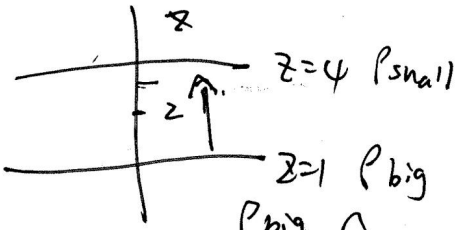
$0 \quad 1$

(c) (4 points) A path drawn through a uniform electric field **E** is divided into four segments of equal length labeled A, B, C, D. The path is traversed in the direction shown.



For which segment there is no change in potential?

(a) Segment A
(b) Segment B
(c) Segment C
(d) Segment D



Because $\vec{E} = \frac{\rho_{big}}{2\varepsilon_0}\hat{z}$

(d) (5 points) Suppose there are two positively charged sheets parallel to the xy-plane at z = 1 and at z = 4. Suppose that the charge density of the z = 1 plane 1 is greater than that of the z = 4 plane. Suppose my starting point is (0, 0, 2) and my finishing point is (1, 1, 3). Which of the following is true?

$\vec{E}_2 = \frac{\rho_{small}}{2\varepsilon}\hat{z}$

(a) There is a (positive) voltage rise travelling between the two points.
(b) There is a (positive) voltage drop travelling between the two points.
(c) There is no voltage change travelling between the two points.

$\therefore E_{total}$ is pointing to $\hat{z}$ direction, in which voltage decrease

(e) (4 points) The charges on a parallel plate capacitor in air are held at fixed amount, when filling it with a dielectric material with a relative permittivity of 10, the voltage _____ and the capacitance _____.

(a) decreases, decreases
(b) decreases, increases
(c) increases, decreases
(d) increases, increases
(e) stays the same, stays the same

$\varepsilon_r = 10$

$C \uparrow = \frac{Q}{V\downarrow}$ $\checkmark$ $C \uparrow = \frac{\varepsilon \uparrow A}{d}$ $\checkmark$

(f) (4 points) Suppose the charge density is zero everywhere inside a three-dimensional homogenous dielectric material that electrostatic conditions apply, what can be concluded.

(a) The displacement flux density is zero
(b) The electric field is zero
(c) The divergence of polarization is zero
(d) The electrostatic potential is constant

$\nabla \cdot D = \rho = 0$

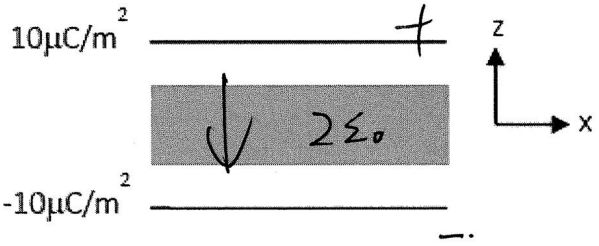
$\nabla \cdot P = 0 = \rho_{bound}$.

because charge density = 0

$\therefore$ bound charge density = 0

$\therefore \rho_{bound} = 0$

(g) (5 points) An infinite plane dielectric slab lies between two infinite plane sheets of uniform charge density of $\pm 10\ \mu C/m^2$. The slab has a permittivity of $2\epsilon_0$. What is the polarization P inside the slab?



(a) 10E-6 $\hat{z}$ (C/m^2)
(b) -10E-6 $\hat{z}$ (C/m^2)
(c) 5E-6 $\hat{z}$ (C/m^2)
(d) -5E-6 $\hat{z}$ (C/m^2)
(e) 2.5E-6 $\hat{z}$ (C/m^2)
(f) -2.5E-6 $\hat{z}$ (C/m^2)

$P = D - \epsilon_0 E$

$E = 2 \times \frac{\rho}{2\epsilon} (-\hat{z})$

$= \frac{\rho}{\epsilon} (-\hat{z})$

$= \frac{10\mu C}{2\epsilon_0} (-\hat{z})$

$\chi_e = \epsilon_r - 1 = 1$

$\therefore D = 5\mu C.$

$P = \chi_e \epsilon_0 E$

$= \epsilon_0 \cdot \frac{5\mu C}{\epsilon_0} = 5 \times 10^{-6} (-\hat{z}) \frac{C}{m^2}$

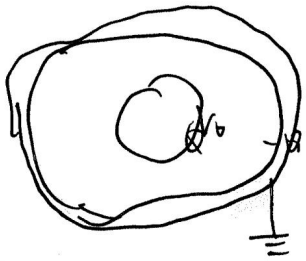
$\chi_e = \epsilon_r - 1$

PART II: DETAILED QUESTIONS

2. A coaxial cable has an inner conductor with radius a and an outer conductor with radius b. The space between the two conductors is filled with air. When the inner conductor is at a potential of V_o with respect to the grounded outer conductor,

   (for a coaxial cable $\mathbf{E} = \frac{Q}{2\pi\epsilon l r}$)

   (a) (5 points) What is the charge per unit length on the inner conductor?



$C = \frac{Q}{V} = \frac{Q}{-\int \vec{E}\,d\vec{l}} = \frac{Q}{-\frac{Q}{2\pi\epsilon l}\ln r\Big]_b^a}$

$C = \frac{\epsilon A}{b-a} \approx \frac{\epsilon A}{\ln\frac{b}{a}} \approx \frac{2\pi}{\ln\frac{b}{a}}\epsilon L$

$C = \frac{Q}{V}$

$Q = CV = \frac{\epsilon A V_0}{(b-a)} \approx \frac{2\pi\epsilon V_0 L}{\ln\frac{b}{a}}\ [C]$

$\therefore \rho = \frac{Q}{A} = \frac{\epsilon V_0}{(b-a)}\ \left[\frac{C}{m^2}\right] \approx \frac{2\pi\epsilon V_0}{\ln\frac{b}{a}}\ \left[\frac{C}{m^2}\right]$

   (b) (5 points) What is the electric field intensity at r=$\frac{a+b}{2}$?

$E = \frac{Q}{2\pi\epsilon l\left(\frac{a+b}{2}\right)}\ \frac{V}{m}$

$= \frac{Q}{2\pi\epsilon l\left(\frac{a+b}{2}\right)}\ \frac{A}{F}\ V\ \left[\frac{1}{m}\right]$

$E = \frac{Q}{2\pi\epsilon l\left(\frac{a+b}{2}\right)}\hat{r} = \frac{Q}{\pi\epsilon l(a+b)}\hat{r}\ \left[\frac{V}{m}\right]$

3. A charge Q sits at the origin. There is an extremely tiny surface of area dA located at (x, y, z) = (0, 0, 2) [m] with its normal vector pointing along the positive z-direction. For this configuration, the displacement flux through this area is 10 C. What would the displacement flux be if the area is moved to the following locations without changing its angular orientation (i.e., without changing the direction of the normal vector)?

(a) (5 points) Moved to (0, 0, -1) m?

$$\int \vec{D} \cdot d\vec{A} = 10\,C$$

$$\int \epsilon_0 \vec{E}_{(0,0,2)} \cdot d\vec{A} = 10\,C.$$

$$\vec{E}_{(0,0,2)} = \frac{kQ}{4}\hat{z}\ \frac{V}{m}$$

$$\vec{E}_{(0,0,-1)} = \frac{kQ}{1}(-\hat{z})\ \frac{V}{m} = -4\,\vec{E}_{(0,0,2)}$$

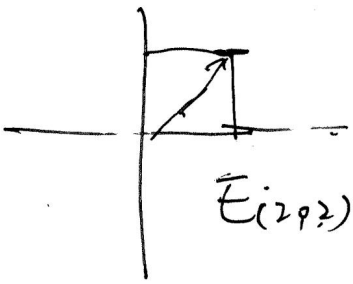
$$\therefore \int \epsilon_0 \vec{E}_{(0,0,-1)} \cdot d\vec{A} = -4 \int \epsilon_0 \vec{E}_{(0,0,2)} \cdot d\vec{A} = -40\,C.$$

(b) (5 points) Moved to (2, 0, 0) m?

Because, the ~~electric~~ direction of $\vec{E}$ in $(2,0,0)$ is points at $\hat{x}$, and the normal vector of the surface points at $\hat{z}$ so $\hat{x} \cdot \hat{z} = 0$

$$\therefore \int \vec{D}_{(2,0,0)} \cdot d\vec{A} = 0\ C$$

(c) (5 points) Moved to (2, 0, 2) m?



$$\vec{E}_{(2,0,2)} = \frac{kQ}{8}\left(\frac{\hat{x}}{\sqrt{2}} + \frac{\hat{z}}{\sqrt{2}}\right)$$

$$\therefore \int \epsilon_0 \vec{E}_{(2,0,2)} \cdot d\vec{A} = \int \epsilon_0 \cdot \frac{kQ}{8}\left(\frac{\hat{x}}{\sqrt{2}} + \frac{\hat{z}}{\sqrt{2}}\right) \hat{z}\, dA$$

$$= \int \epsilon_0 \frac{kQ}{8\sqrt{2}}\, dA$$

$$= \frac{1}{2\sqrt{2}} \int \epsilon_0 \frac{kQ}{4}\, dA = \frac{10}{2\sqrt{2}} = \frac{5\sqrt{2}}{2}\,C.$$

4. The electric field in free space is given by the equation $\mathbf{E} = < x^2 z, 2yz, f(x,y,z) >$ [V/m]. The volumetric charge density is zero everywhere.

(a) (10 points) Find a possible function for f.

$\nabla D = 0 \quad \nabla \cdot \epsilon_0 E = 0$

$2xz\epsilon_0 + 2z\epsilon_0 + \frac{\partial}{\partial z} f(x,y,z)\epsilon_0 = 0$ ∵ free space → ∴ ϵ_0 homogenous

$\frac{\partial}{\partial z} f(x,y,z) = -2xz - 2z$

$f(x,y,z) = \int -2xz - 2z \, dz$

$= -xz^2 - z^2 + C.$

When $C=0$ $f(x,y,z) = -xz^2 - z^2$

(b) (10 points) For the function f you found, show whether or not this is an electrostatic field?

$E = < x^2 z, \quad 2yz, \quad -xz^2 - z^2 >$

$\nabla \times E = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 z & 2yz \cdot -xz^2 - z^2 & \end{vmatrix}$

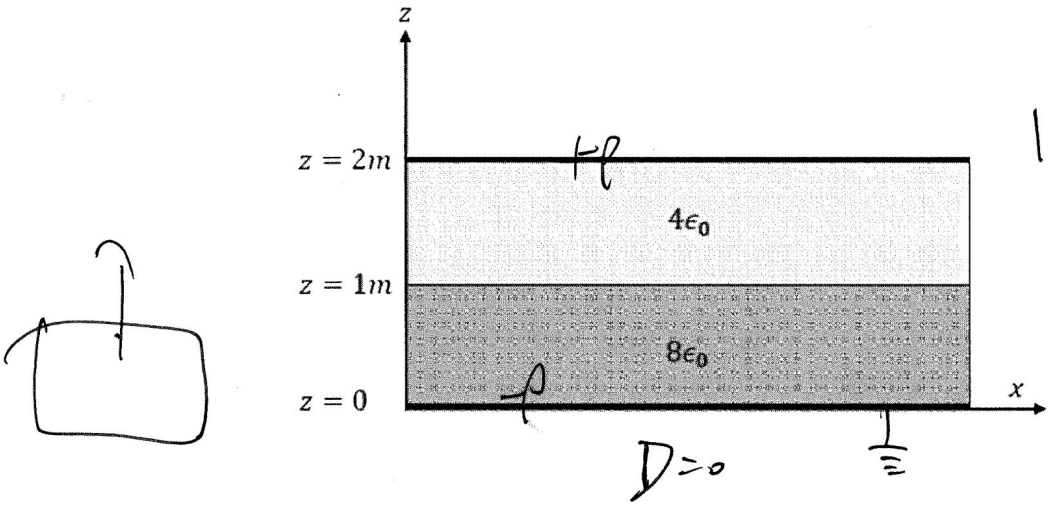
$= \hat{x}(0 - 2y) - \hat{y}(-z^2 - x^2) + \hat{z}(0 - 0)$

$= -2y\hat{x} + (x^2 + z^2)\hat{y}$

≠ 0 Not always = 0 for all x, y, z

∴ Not electrostatic

5. Consider two infinite planar dielectric slabs, each with a thickness of 1m but different dielectric constants (i.e., relative permittivity), laying parallel to the xy-plane in free space (i.e., vacuum). The bottom slab has a dielectric constant ϵ_r=8, and the top slab has a dielectric constant ϵ_r=4. Place two infinitely large, zero thickness, conducting plates at z=0m and z=2m to form a capacitor. The plate at z=0m is grounded, and the plate at z=2m is held at 1V. Please answer the following questions. Note that for multiple choice questions, **there could be more than one correct answer.**



(a) (5 points) What is the functional form of V in each segment, where a and b are constants that may have different values in the two different slabs?

(a) V=a
(b) V=a*z+b
(c) V=a*z^2+b
(d) V=a*sin(z)

(b) (5 points) What is/are the boundary condition(s) at z=0m?

(a) $D_{z+} \neq D_{z-}$
(b) $D_z = 0$
(c) $E_{z+} = E_{z-}$
(d) $E_z = 0$
(e) V=0
(f) V=1

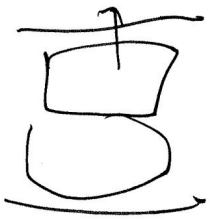
D=ε₀E

(c) (5 points) What is/are the boundary condition(s) at z=1m?

(a) $D_{z+} = D_{z-}$
(b) $D_z \neq 0$
(c) $E_{z+} = E_{z-}$
(d) $E_z \neq 0$
(e) $V_{z+} = V_{z-}$

(d) (5 points) What is/are the boundary condition(s) at z=2m?

(a) $D_{z+} = D_{z-}$
(b) $D_z = 0$
(c) $E_{z+} = E_{z-}$
(d) $E_z = 0$
(e) V=0
(f) V=1



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